HOW COGNITIVE BIASES AFFECT XAI-ASSISTED DECISION-MAKING: A SYSTEMATIC REVIEW

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Paper presentation

INTRODUCTION

Overview

- **What - XAI:** Brings transparency to AI.
- **Why - Problem:** Cognitive biases could affect explainability systems.
- **How - Work:** Create a general vision of the problem ← 37 papers, 53 biases
- **Where - Aim:** XAI aligns with people's cognitive process.

BACKGROUND

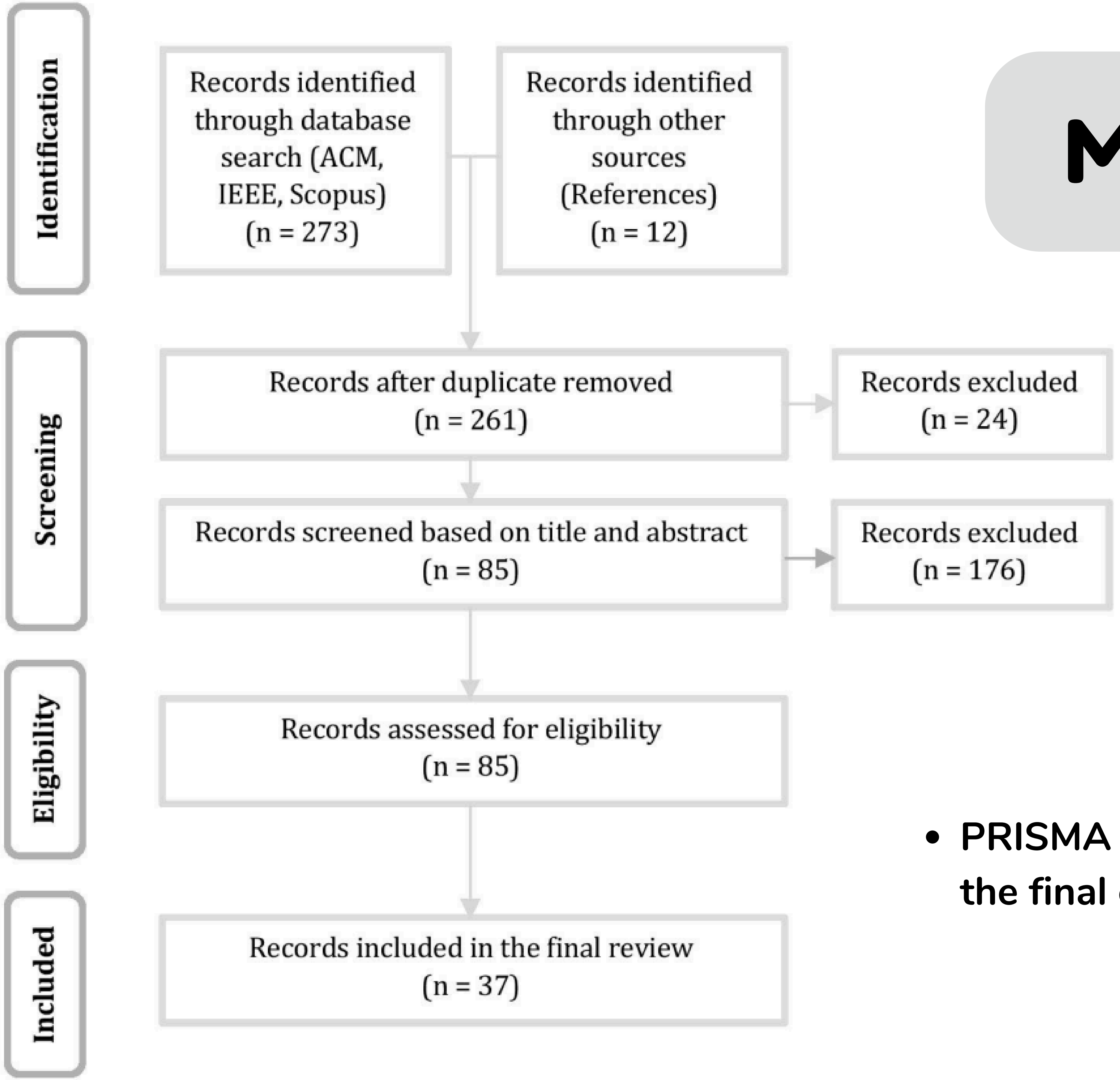
Cognitive bias & Research Gap → 4 Research Questions

Cognitive Biases
- Unjustified trust in AI
- Reasoning errors
- Confirmation bias

Research Gap
- Only ruled-based explanation
- Doesn’t cover all the biases

4 Research Questions
- Biases studied
- Bias contexts
- Mitigate vs. leverage
- Future challenges

METHODOLOGY

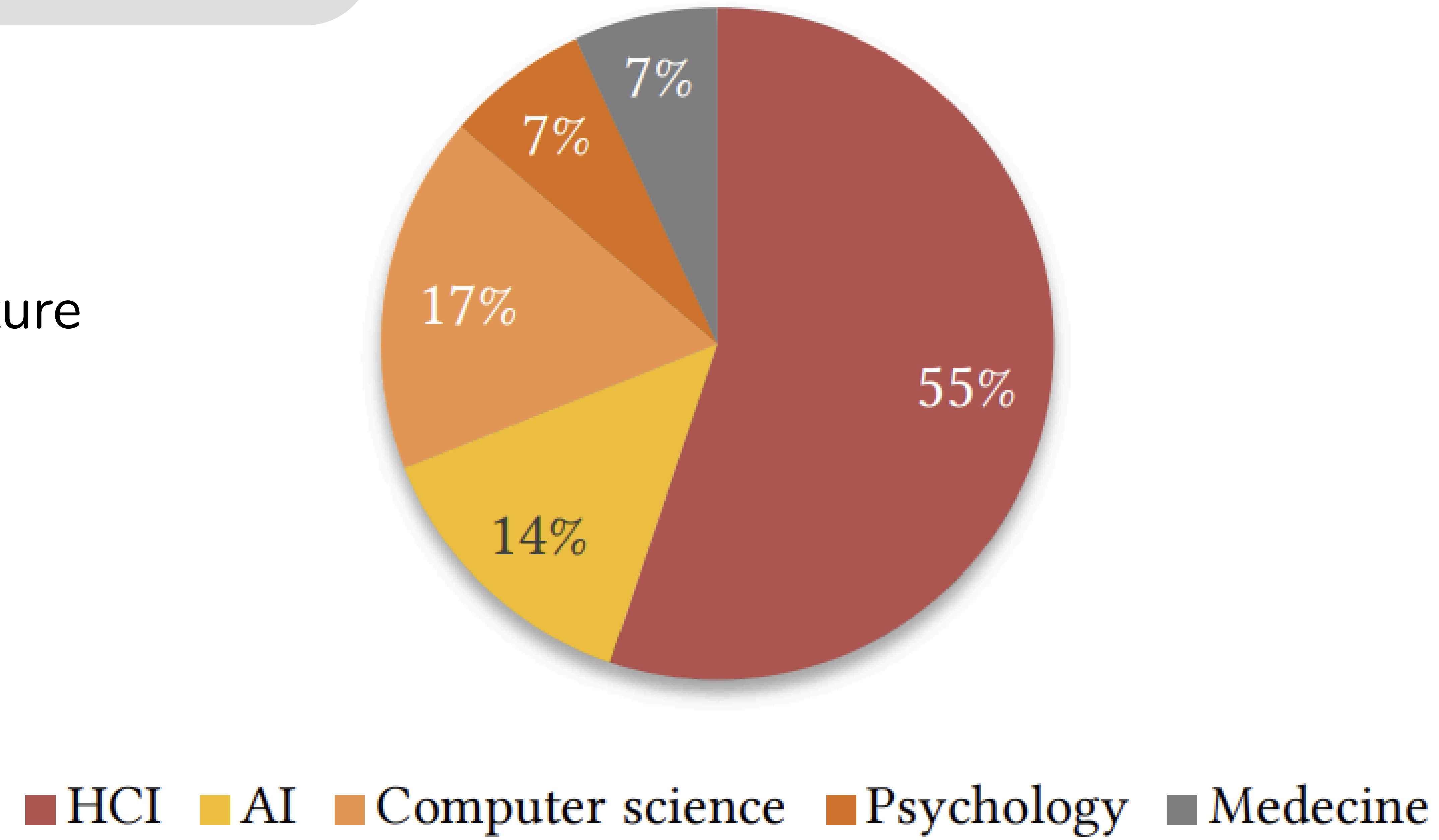


- **PRISMA flow diagram [21] on how the final corpus was curated**

METHODOLOGY

In the corpus of 37 papers ->

- 7 papers are reviews of the literature
- 29 papers are primary studies



Distribution of the corpus across the disciplines, showing the diversity of the subject areas.

RESULTS

Four Ways Cognitive Biases Interact with XAI

Design Constraints

- Prefer simple, complete explanations
- Focus on abnormal/intentional causes
- Expect social interaction

Evaluation Biases

- Preferences ≠ performance
- More attention to false negatives
- Change blindness in visuals

Mitigated Biases

- Pre-use algorithmic optimism
- Representativeness bias
- Misattributed trust
- Belief in unsupported claims

Exacerbated Biases

MAJOR CONCERN

- Overreliance on AI
- False causal narratives
- Under-reliance on AI

RESULTS

53

Biases identified

4

Successfully Mitigated

15+

Made worse

ADDITIONAL RESULTS

Corpus Breakdown
285 papers screened / 37 reviewed mostly from Human-Computer Interaction research

Cognitive biases vary across stakeholders, suggesting **tailored XAI designs** for users, developers, and regulators.

7 Research directions focus on bias harm, prioritization, attention metrics, social behavior, explanation timing, autonomy, and accountability.

DISCUSSIONS

Biases are natural heuristics, not just errors.

Different users experience biases differently

Use qualitative metrics like attention tracking.

Empower users with tools to question AI.

LIMITATIONS

Search Strategy Constraints

- Keyword based approach may have missed articles using specific **biased terminology**
- Limited to **ACM, IEEE**, and **Scopus databases**

Author Bias

- **Interpretation influenced** by researchers' disciplinary backgrounds
- Potential **selection bias** in manual paper additions

Scope & Completeness

- **200+** other cognitive biases remain unexplored in **XAI** context
- **53 biases** identified doesn’t represent the complete universe of relevant cognitive biases

FUTURE WORK

Specific domain exploration
(medical, legal, financial XAI)

Standardized evaluation protocols for bias detection

Interactive versions of a **heuristic map** to facilitate traking of cognitive biases

CONCLUSION

- **53 cognitive biases** identified across **XAI literature**
- **Four way classification** of bias **XAI interactions**
- **Human centered XAI** requires **cognitive science integration**
- Systematic identification of **five key areas** where the **XAI** field needs focused work to better handle **cognitive biases**

Ultimate goal is to make XAI systems aligned with human cognitive processes

THANK YOU

Presentation by Harper Russo